

48-Hour AI Engineering Challenge

You have been selected to complete the Magna Labs technical challenge. This document contains everything you need to know — what to build, how to submit, and what we are looking for.

48 Hours

From first login

Solo

Individual only

Live URL

Deployed app required

Video

Walkthrough required

WHAT YOU ARE BUILDING

AI Content Marketing Suite

[Marketing & Content Domain](#)

Build a production-ready SaaS web application that helps marketers create, manage, and optimize content using AI. The product must feel like a real tool — clean interface, solid backend APIs, and intelligent features throughout. Users can generate written content for multiple formats and generate a matching AI image for each post, all in one seamless workflow.

REQUIRED FEATURES

01 AI Content Generator

Users input a topic, tone, target audience, and content type. The app generates polished, ready-to-use content via an LLM. Must support at least 3 content types (blog post, LinkedIn post, ad copy, email) with a distinct prompt strategy per type.

[\[LLM Integration\]](#) [\[Prompt Engineering\]](#) [\[Multi-format\]](#)

02 AI Image Generator — Per Post ★ Featured

After content is generated, the user clicks once to generate a matching AI image. The app auto-builds a visual prompt from the content topic and tone, calls an image API (DALL-E 3, Stability AI, or similar), and displays the image alongside the text. Users can regenerate with a different style.

[\[Image Generation API\]](#) [\[Auto Prompt Builder\]](#) [\[Content-Visual Pairing\]](#)

03 Content History & Dashboard

All generated content and images are saved per session. Users can view, copy, download, and delete past generations. Each entry shows the text and its image. Dashboard must be clean, paginated, and fast.

[\[Frontend & UI/UX\]](#) [\[Backend API\]](#) [\[Data Persistence\]](#)

04 AI Content Improver

Users paste existing text and choose an improvement goal: shorter, more persuasive, more formal, SEO-optimized, or rewritten for a different audience. The LLM returns a refined version with a short explanation of what changed.

[\[Prompt Chaining\]](#) [\[UX Design\]](#)

05 REST API Backend

All features — text generation, image generation, content storage — must be powered by clean REST API endpoints. No direct LLM or image API calls from the frontend. All AI logic is server-side. Endpoints documented in the README.

[\[Backend Architecture\]](#) [\[API Design\]](#) [\[Security\]](#)

06 Claude Code Workflow

Use Claude Code as your primary development tool throughout the build. Show this in your video: architecture planning, debugging sessions, refactoring, or prompt-driven code generation.

[\[Claude Code\]](#) [\[AI-Native Dev\]](#)

HOW TO ACCESS THE CHALLENGE

1 Visit the Portal Go to the challenge portal URL you have been given by Magna Labs.	2 Enter Your Email Type the email address that was registered for you. If your email is not in the system, access will be denied.	3 Timer Starts Your 48-hour countdown begins the moment you log in for the first time. It will not reset.	4 Submit Your Work Before the timer expires, submit your live app URL and your video walkthrough link through the portal.
--	---	---	---

HOW YOUR WORK IS SCORED

25 pts LLM & Prompt Quality Output quality, prompt strategy per content type, relevance and coherence	20 pts AI Image Generation Auto prompt quality, image relevance to content, smooth UX flow	20 pts Backend & API Design Structure, error handling, security, README documentation
15 pts Frontend & UI/UX Usability, design quality, responsiveness, text and image layout	15 pts Claude Code Usage Demonstrated AI-native workflow shown clearly in your video	+10 pts Bonus Brand voice settings, image style picker, export to PDF or doc

WHAT TO SUBMIT

REQUIRED Live App URL Your deployed web application, publicly accessible. Vercel, Railway, Render, or any platform. Must stay live for 7 days after submission.	REQUIRED Video Walkthrough (5–10 min) Screen recording of your product working end-to-end. Show the text generation flow, the image generation flow, and a portion of your Claude Code workflow.
REQUIRED GitHub Repository Public repo with a clean README: setup steps, tech stack, API docs, and a brief note on architecture decisions.	OPTIONAL Architecture Note A short document (1 page max) explaining your design choices, trade-offs made, and what you would build next if you had more time.

RULES

- You may use any LLM API, image generation API, framework, or database of your choice.
- AI coding assistants are encouraged — Claude Code, Cursor, GitHub Copilot, etc.
- Your video must clearly show the text generation and image generation flows working end-to-end.
- Third-party UI component libraries are allowed. Starting from a full boilerplate template is not.
- Your deployed URL must remain publicly accessible for at least 7 days after submission.
- The 48-hour timer starts when you first log in to the challenge portal. It does not reset.
- This is a solo assessment. No team submissions.

Ready when you are.

Log in to the portal with your registered email address to begin. Your 48-hour countdown starts the moment you log in. Submit your live app URL and your video through the portal before the timer expires.

Good luck. — Magna Labs Hiring Team